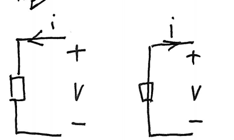
chapter1

$i=\frac{dq}{dt}$ , $1.602\times10^{-19}C$/electron

$v=\frac{dw}{dq}$ , $p=\frac{dw}{dt}=\frac{dw}{dq}\frac{dq}{dt}=vi$

passive sign convention

enter through the terminal is more positive.

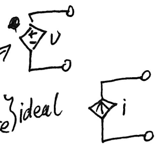



absorb supply

active elements: Generators, Batteries, OA

passive elements: Resistors, capacitors, Inductors

Voltage or current source (independent or dependent)




Voltage source: no resistance
current source: infinite resistance } ideal

voltage - controlled voltage source
V C V S

chapter3

nodal analysis KCL
{ ① select a node as reference node. Assign $V_1 \cdots V_{n-1}$ to other $N-1$ node
② Apply KCL, use ohm's law to express the branch current in terms of node voltage
③ solve the remaining $n-1$ simultaneous equation to obtain the unknow node voltages

supernode: KCL apply to a closed boundary

mesh analysis KVL
{ ① Assign mesh current $i_1, i_2 \cdots$ in to the n meshes
② Apply KVL of each of the n meshes, use ohm's law to express the voltage in terms of the mesh current.
solve the resulting n simultaneous equation to get mesh currents

Transistor circuits


$I_E=I_B+I_C$

$V_{CE}+V_{EB}+V_{BC}=0$

active, cutoff, saturation
放大 截止 饱和

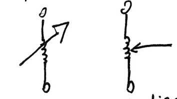
$I_C=\alpha I_E$, $I_C=\beta I_B$, $\beta=\frac{\alpha}{1-\alpha}$

chapter2

$R=\rho\frac{L}{A}$ , $G=\frac{1}{R}$ (电导) siemens(S) Mhos (℧)

$V=iR$

short circuit (短路)
open circuit (开路)



variable Resistor    potentiometer 分压器

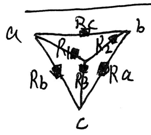
Ⓥ voltmeter

series: $R_{eq}=\sum_{n=1}^{N}R_n$

parallel: $\frac{1}{R_{eq}}=\sum_{n=1}^{N}\frac{1}{R_n}$ , $G_{eq}=\sum_{i=1}^{N}G_i$

KCL
close boundary
$\sum_{n=1}^{N}i_n=0$

KVL
Loop
$\sum_{m=1}^{M}i_m=0$




Δ→Y

$R_1=\frac{R_bR_c}{R_a+R_b+R_c}$

$R_2=\frac{R_cR_a}{R_a+R_b+R_c}$

$R_3=\frac{R_aR_b}{R_a+R_b+R_c}$

$R_a=\frac{R_1R_2+R_2R_3+R_3R_1}{R_1}$

$R_b=\frac{R_1R_2+R_2R_3+R_3R_1}{R_2}$

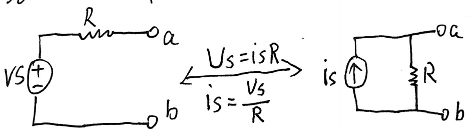
$R_c=\frac{R_1R_2+R_2R_3+R_3R_1}{R_3}$

chapter4

linearity: $kiR=kV$
$V=(i_1+i_2)R=i_1R+i_2R=V_1+V_2$

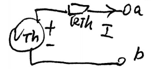
superposition: ① Turn off all independent sources except one source, by replacing every voltage source by 0V (short circuit), current source by 0A (or open circuit). Find output due to that active source.
② Repeat step 1 for each of the other independent sources.
③. Find the total contribution by adding algebraically all the contributions due to the indepent sources

sources transformation




$V_S=i_sR$
$i_s=\frac{V_S}{R}$

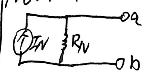
Thevenin theorem




Vth: open circuit voltage

Rth: input or equivalent resistance measured at the terminals when all the independent sources are turned off.

Norton theorem




$R_N$: same to Rth

$I_N$: short cut current

Maximum power transfer

$P=i^2R_L=\left(\frac{V_{Th}}{R_{Th}+R_L}\right)^2R_L$

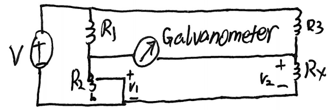
$P_{max}=\frac{V_{Th}^2}{4R_{Th}}$ , $R_L=R_{Th}$

Real source

$V_L=\frac{R_L}{R_S+R_L}V_S$

$i_L=\frac{R_P}{R_P+R_L}i_s$

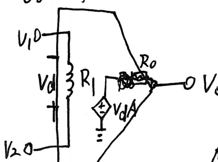
wheatstone bridge

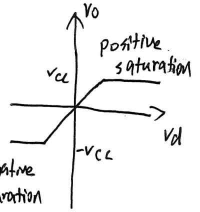


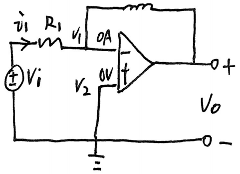

$R_2R_3=R_1R_X$

chapter5 OA

$V_o=AV_d=A(V_2-V_1)$



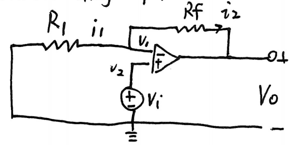




inverting Amplifier

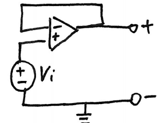
$V_o=-\frac{R_f}{R_i}V_i$

noninverting amplifier



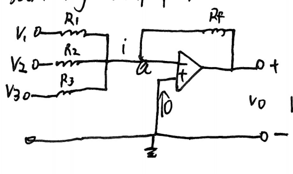

$V_o=(1+\frac{R_f}{R_1})V_i$

voltage follower   isolute, $R_i:\infty$, $R_o:0$



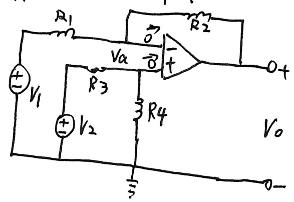

$V_o=V_i$

Summing Amplifier



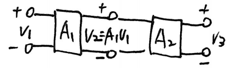

$V_o=-\left(\frac{R_f}{R_1}V_1+\frac{R_f}{R_2}V_2+\frac{R_f}{R_3}V_3\right)$

Difference Amplifier




$V_o=\frac{R_2(1+R_1/R_2)}{R_1(1+R_3/R_4)}V_2-\frac{R_2}{R_1}V_1$

cascaded Op Amp




$V_3=A_2V_2$ - - -

$A=A_1A_2\cdots$

chapter6 $q=CV$ , $C=\frac{\varepsilon A}{d}$ , $i=\frac{dq}{dt}=C\frac{dv}{dt}$ , $v(t)=\int_{t_0}^{t}\frac{1}{C}i(\tau)d\tau+v(t_0)$

$p=vi=Cv\frac{dv}{dt}$ , $w=\int_{t_0}^{t}Cv\frac{dv}{d\tau}d\tau=C\int_0^v v\,dv=\frac{1}{2}Cv^2=\frac{q^2}{2C}$

Parallel: $C_{eq}=\sum_{i=1}^{N}C_i$

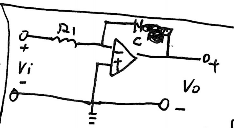
series: $\frac{1}{C_{eq}}=\sum_{i=1}^{M}\frac{1}{C_i}$

$V=L\frac{di}{dt}$, $L=\frac{N^2\mu A}{l}$ , $\oint\vec{B}\cdot d\vec{s}=\mu_0 I_{enc}$ , $V=L\frac{di}{dt}\Rightarrow di=\frac{1}{L}vdt$

$i(t)=\frac{1}{L}\int_{t_0}^{t}v(\tau)d\tau+i(t_0)$ , $P=vi=Li\frac{di}{dt}$ , $w=\int_{t_0}^{t}(Li\frac{di'}{dt})d\tau=\int_0^i Li\,di'$
$=\frac{1}{2}Li^2$

series: $L_{eq}=\sum_{i=1}^{N}L_i$

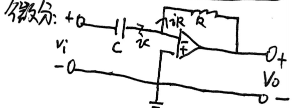
parallel: $\frac{1}{L_{eq}}=\sum_{i=1}^{M}\frac{1}{L_i}$




积分

$V_o=-\frac{1}{RC}\int_0^t V_i(\tau)d\tau$

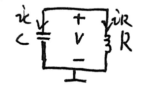
微分:




$V_o=-RC\frac{dV_i}{dt}$

chapter7 $i_C+i_R=0$

RC




$i_C+i_R=0$ voltage: $V(t)=V_0e^{-t/RC}$ , $\tau=RC$

$\frac{cdv}{dt}+\frac{v}{R}=0$  $V(t)=V_0e^{-t/\tau}$ , 5 five time constants

current: $i_R(t)=\frac{v(t)}{R}=\frac{V_0}{R}e^{-t/\tau}$

power dissipated in the resistor

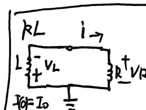
$P(t)=Vi_R=\frac{V_0^2}{R}e^{-2t/\tau}$

energy - - - -

$W_R(t)=\frac{1}{2}CV_0^2(1-e^{-2t/\tau})$

从 C 两端 去找 Rth

RL




$V_L+V_R=0$

$L\frac{di}{dt}+Ri=0$

$i(t)=I_0e^{-Rt/L}$

$\tau=\frac{L}{R}$

$V_R(t)=I_0Re^{-t/\tau}$

$P_R(t)=I_0^2Re^{-2t/\tau}$

$W_R=\frac{1}{2}LI_0^2(1-e^{-2t/\tau})$

stepfunction

$u(t)=\begin{cases}0 & t<0\\1 & t>0\end{cases}$

$\int_{0^-}^{0^+}\delta(t)dt=1$ , $\delta(t)=\frac{du}{dt}$ , $u=\int_{-\infty}^{t}\delta(t)dt$

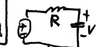
$r(t)=\begin{cases}0 & t\le0\\t & t\ge0\end{cases}$ , $u=\frac{dr}{dt}$ , $r=\int_{-\infty}^{t}u(\lambda)d\lambda$

$u(t-t_0)$, $t_0>0$, delayed by $t_0$

$u(t+t_0)$, $t_0<0$, advanced by $|t_0|$

$\tau=\frac{L}{R}$

step response RC $\tau=RC$




$\frac{cdv}{dt}+\frac{v-V_s}{R}=0$

$V(t)=\begin{cases}V_0 & t<0\\V_s+(V_0-V_s)e^{-t/\tau} & t>0\end{cases}$

$V_n=V_0e^{-t/\tau}$

$V_f=V_s(1-e^{-t/\tau})$

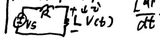
$V(t)=V(\infty)+[V(t_0)-V(\infty)]e^{-(t-t_0)/\tau}$

$i(t)=C\frac{dv}{dt}=\frac{V_s}{R}e^{-t/\tau}u(t)$

$V(t)=L\frac{di}{dt}=V_se^{-t/\tau}u(t)$

瞬时值 = 最终值 + [初始值 − 最终值] $\cdot e^{-(t-t_0)/\tau}$

RL step response




$L\frac{di}{dt}+iR=V_s\Rightarrow i(t)=\begin{cases}I_0\\ \frac{V_s}{R}+(I_0-\frac{V_s}{R})e^{-t/\tau}\end{cases}$

$i(t)=i(\infty)+[i(t_0)-i(\infty)]e^{-(t-t_0)/\tau}$

## second-order circuit

Source-Free series RLC circuit

$Ri + L\frac{di}{dt} + \frac{1}{C}\int_{-\infty}^{t} i(\tau)d\tau = 0$

$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0$ $\begin{cases} i(0) = I_0 \\ \frac{di(0)}{dt} = -\frac{1}{L}(RI_0 + V_0) \end{cases}$

特征方程, $s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$

$s^2 + 2\alpha s + \omega_0^2 = 0$

$\alpha = \frac{R}{2L}$ $\quad S = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$

$\omega_0 = \frac{1}{\sqrt{LC}}$

α: damping factor

$\omega_0$: undamped natural frequency

$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ ($\alpha > \omega_0$) over

$i(t) = e^{-\alpha t}(A_1 + A_2 t)e^{-\alpha t}$ ($\alpha = \omega_0$) critical

$i(t) = e^{-\alpha t}(A_1 \cos\omega_d t + A_2 \sin\omega_d t)$ under

$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ ($\omega_0 > \alpha$)

If $\alpha = 0$, ⇒ undamped

source-free parallel RLC circuit

$\frac{V}{R} + C\frac{dv}{dt} + \frac{1}{L}\int_{-\infty}^{t} v(\tau)d\tau = 0$

$\frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{1}{LC}v(t) = 0$

$\begin{cases} V(0) = V_0 \\ \frac{dV(0)}{dt} = -\frac{1}{RC}(V_0 + RI_0) \end{cases}$

特征方程, $s^2 + 2\alpha s + \omega_0^2 = 0$

$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0$

$\alpha = \frac{1}{2RC}$ $\quad S = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$

$\omega_0 = \frac{1}{\sqrt{LC}}$ $\quad V(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ ($\alpha > \omega_0$)

$V(t) = e^{-\alpha t}(A_1 + A_2 t)$ ($\alpha = \omega_0$)

$V(t) = e^{-\alpha t}(A_1 \cos\omega_d t + A_2 \sin\omega_d t)$

$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$, If $R = \infty$, $\alpha = 0$, undamped

step response of series RLC

$L\frac{di}{dt} + Ri + v = V_s$, $i = C\frac{dv}{dt}$

$\frac{d^2v}{dt^2} + \frac{R}{L}\frac{dv}{dt} + \frac{v}{LC} = \frac{V_s}{LC}$

$\alpha = \frac{R}{2L}$ $\quad S = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$

$\omega_0 = \frac{1}{\sqrt{LC}}$ $\quad v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$ over

$v(t) = V_s + (A_1 + A_2 t)e^{-\alpha t}$ critical

$v(t) = V_s + (A_1 \cos\omega_d t + A_2 \sin\omega_d t)e^{-\alpha t}$ under

$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ $\quad v(t) = v_t + v_{ss}$

~~source~~ step response of parallel RLC

$\frac{V}{R} + i + C\frac{dv}{dt} = I_s$, $V = L\frac{di}{dt}$

$\frac{d^2i}{dt^2} + \frac{1}{RC}\frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC}$

$\alpha = \frac{1}{2RC}$ $\quad \omega_0 = \frac{1}{\sqrt{LC}}$

$S = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$

$i(t) = I_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$ over

$i(t) = I_s + (A_1 + A_2 t)e^{-\alpha t}$ critical

$i(t) = I_s + (A_1 \cos\omega_d t + A_2 \sin\omega_d t)e^{-\alpha t}$ underdamped

$i(t) = i_t(t) + i_{ss}(t)$

$i_{ss}(t)$ = steady state part

General second-order

① find $x(0)$, $dx(0)/dt$

② KVL, KCL ⇒ ODE ⇒ 特征根 ⇒ $x_t(t)$

③ $x_{ss}(t) = x(\infty)$

④ $x(t) = x_t(t) + x_{ss}(t)$

Sinusoids and phasors

$v(t) = V_m \sin(\omega t + \phi)$

phasors: complex number:

$z = x + jy$, $z = r\angle\phi$, $z = re^{j\phi}$

$r = \sqrt{x^2 + y^2}$, $\phi = \tan^{-1}\frac{y}{x}$

$z = x + jy = r\angle\phi = re^{j\phi} = r(\cos\phi + j\sin\phi)$

$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$

$z_1 z_2 = r_1 r_2 \angle\phi_1 + \phi_2$

$z_1/z_2 = (r_1/r_2)\angle\phi_1 - \phi_2$, $\frac{1}{z} = \frac{1}{r}\angle -\phi$

$\sqrt{z} = \sqrt{r}\angle\phi/2$, $z^* = x - jy$

$V_m\cos(\omega t + \phi) = r\angle\phi$

$V_m\angle\phi = re^{-j\phi}$

$V_m \sin(\omega t + \phi) \Rightarrow V_m\angle\phi - 90°$

| Time domain | frequency domain |
|---|---|
| $\frac{dv}{dt}$ | $\Leftrightarrow j\omega V$ |
| $\int v dt$ | $\Leftrightarrow \frac{V}{j\omega}$ |

$i = I_m\cos(\omega t + \phi) \Rightarrow I_m\angle\phi = I$

$v = RI_m\angle\phi \Rightarrow V = RI$

$v = L\frac{di}{dt} \Rightarrow V = j\omega LI$

$i = C\frac{dv}{dt}$, $V = \frac{I}{j\omega C}$

$V = ZI$, $Z = R + jX$

$Z = |Z|\angle\theta$, $|Z| = \sqrt{R^2 + X^2}$, $\theta = \tan^{-1}\frac{X}{R}$

| Element | Impedance | Admitance |
|---|---|---|
| R | $Z = R$ | |
| L | $Z = j\omega L$ | $Y = \frac{1}{Z}$ |
| C | $Z = 1/j\omega C$ | |

$R = \frac{V}{I} = \frac{V_m}{I_m}\angle\theta_v - \theta_i$ (load impedance)

## source Transformation

$V_s = Z_s I_s \Leftrightarrow I_s = \frac{V_s}{Z_s}$

$V_{Th} = Z_N I_N$

$Z_{Th} = Z_N$

瞬时功率: $P = VI$

$V(t) = V_m\cos(\omega t + \theta_v)$

$I(t) = I_m\cos(\omega t + \theta_i)$

$P(t) = VI = \frac{1}{2}V_m I_m\cos(\theta_v - \theta_i) + \frac{1}{2}V_m I_m\cos(2\omega t + \theta_v + \theta_i)$

平均功率: $P = \frac{1}{T}\int_0^T P(t)dt$

$P = \frac{1}{2}V_m I_m\cos(\theta_v - \theta_i)$

$V = V_m\angle\theta_v$, $I = I_m\angle\theta_i$

$\frac{1}{2}VI^* = \frac{1}{2}V_m I_m\angle\theta_v - \theta_i$

$= \frac{1}{2}V_m I_m[\cos(\theta_v - \theta_i) + j\sin(\theta_v - \theta_i)]$

$P = \frac{1}{2}\mathrm{Re}\{RI^*\}$

If $\theta_v = \theta_i$, $P = \frac{1}{2}V_m I_m = \frac{1}{2}I_m^2 R = \frac{1}{2}|I|^2 R$

$\theta_v - \theta_i = \pm 90°$, $P = \frac{1}{2}V_m I_m\cos 90° = 0$

最大平均功率传输

$Z_{Th} = R_{Th} + jX_{Th}$

$Z_L = R_L + jX_L$

$Z_L = Z_{Th}^* \Rightarrow \max$

$P_{max} = \frac{|V_{Th}|^2}{8R_{Th}}$

effective value and Rms value

$P = \frac{1}{T}\int_0^T i^2 R dt = \frac{R}{T}\int_0^T i^2 dt = RI_{eff}^2$

$X_{effective} = X_{RMS} = \sqrt{\frac{1}{T}\int_0^T x^2 dt}$

$Y = Y_m\cos\omega t$

~~$X = X_m\cos\omega t$~~

$Y_{rms} = \frac{Y_m}{\sqrt{2}}$

视在功率: $S = V_{rms}I_{rms} = |S|$

平均功率:

$P = \frac{1}{2}V_m I_m\cos(\theta_v - \theta_i)$

Let $\frac{V_m}{\sqrt{2}} = V_{rms}$, $\frac{I_m}{\sqrt{2}} = I_{rms}$

$\therefore P = V_{rms}I_{rms}\cos(\theta_v - \theta_i)$

$\therefore S = V_{rms}I_{rms}$

Power factor angle: $\theta_v - \theta_i$

power factor: $\cos(\theta_v - \theta_i) = \frac{P}{S}$

复功率 $S = \frac{1}{2}V_m I^*$

$= V_{rms}I_{rms}^*$

$S = P + jQ$ $\quad = \frac{V}{\sqrt{2}}\frac{I^*}{\sqrt{2}}$

$S = \sqrt{P^2 + Q^2}$ $\quad = (V_{rms}\angle\theta_v)(I_{rms}\angle\theta_i)$

$S = V_{rms}I_{rms}\angle\theta_v - \theta_i$

$P = \mathrm{Re}\{S\} = I_{rms}^2 R = V_{rms}\cos(\theta_v - \theta_i)$

$Q = \mathrm{Im}\{S\} = I_{rms}^2 X = I_{rms}\sin(\theta_v - \theta_i)$

$Z = R + jX$

| | | |
|---|---|---|
| $Q = 0$ | resistive | unity pf |
| $Q < 0$ | capacitive | leading pf |
| $Q > 0$ | inductive | lagging pf |

S, Q, P; |Z|, X, R 容性 电压滞后电流 感性 电压超前电流

三相电

positive sequence

negative

手拉预载:

let $\Delta = Y$

$Z_\Delta = 3Z_Y$

| Y-Y | |
|---|---|
| $V_{an} = V_p\angle 0°$ | $V_{ab} = \sqrt{3}V_p\angle 30°$ |
| $V_{bn} = V_p\angle -120°$ | $V_{bc} = V_{ab}\angle -120°$ |
| $V_{cn} = V_p\angle +120°$ | $V_{ca} = V_{ab}\angle +120°$ |
| Same as line currents | $I_a = V_{an}/Z_Y$ |
| | $I_b = I_a\angle -120°$ |
| | $I_c = I_a\angle +120°$ |

| Y-Δ | |
|---|---|
| $V_{an} = V_p\angle 0°$ | $V_{ab} = V_{AB} = \sqrt{3}V_p\angle 30°$ |
| $V_{bn} = V_p\angle -120°$ | $V_{bc} = V_{BC} = V_{ab}\angle -120°$ |
| $V_{cn} = V_p\angle +120°$ | $V_{ca} = V_{CA} = V_{ab}\angle +120°$ |
| $I_{AB} = V_{AB}/Z_\Delta$ | $I_a = I_{AB}\sqrt{3}\angle -30°$ |
| $I_{BC} = V_{BC}/Z_\Delta$ | $I_b = I_a\angle -120°$ |
| $I_{CA} = V_{CA}/Z_\Delta$ | $I_c = I_a\angle +120°$ |
| | Same as phase voltages |

| Δ-Δ | |
|---|---|
| $V_{ab} = V_p\angle 0°$ | |
| $V_{bc} = V_p\angle -120°$ | |
| $V_{ca} = V_p\angle +120°$ | |
| $I_{AB} = V_{ab}/Z_\Delta$ | $I_a = I_{AB}\sqrt{3}\angle -30°$ |
| $I_{BC} = V_{bc}/Z_\Delta$ | $I_b = I_a\angle -120°$ |
| $I_{CA} = V_{ca}/Z_\Delta$ | $I_c = I_a\angle +120°$ |
| Same as phase voltages | Same as phase voltages |

| Δ-Y | |
|---|---|
| $V_{ab} = V_p\angle 0°$ | $I_a = \frac{V_p\angle -30°}{\sqrt{3}Z_Y}$ |
| $V_{bc} = V_p\angle -120°$ | $I_b = I_a\angle -120°$ |
| $V_{ca} = V_p\angle +120°$ | $I_c = I_a\angle +120°$ |
| Same as line currents | |

The Fourier transform of the input voltage is

$$V_i(\omega) = \frac{2}{3 + j\omega}$$

and the transfer function obtained by voltage division is

$$H(\omega) = \frac{V_o(\omega)}{V_i(\omega)} = \frac{1/j\omega}{2 + 1/j\omega} = \frac{1}{1 + j2\omega}$$

Hence,

$$V_o(\omega) = V_i(\omega)H(\omega) = \frac{2}{(3 + j\omega)(1 + j2\omega)}$$

or

$$V_o(\omega) = \frac{1}{(3 + j\omega)(0.5 + j\omega)}$$

By partial fractions,

$$V_o(\omega) = \frac{-0.4}{3 + j\omega} + \frac{0.4}{0.5 + j\omega}$$

Taking the inverse Fourier transform yields

$$v_o(t) = 0.4(e^{-0.5t} - e^{-3t})u(t)$$

## Magnetically coupled circuits

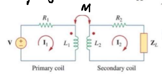
$w = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 \pm M i_1 i_2$ 个同进出Dot +

$M \leq \sqrt{L_1 L_2}$ $\quad k = \frac{M}{\sqrt{L_1 L_2}}$

perfect $k = 1$

losely $k < 0.5$

tightly $k > 0.5$



$V = (R_1 + j\omega L_1)I_1 - j\omega M I_2$

$0 = -j\omega M I_1 + (R_2 + j\omega L_2 + Z_L)I_2$

$Z_{in} = \frac{V}{I_1} = R_1 + j\omega L_1 + \frac{\omega^2 M^2}{R_2 + j\omega L_2 + Z_L}$

$Z_R$, 反射阻抗

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} \frac{L_2}{j\omega(L_1L_2 - M^2)} & \frac{-M}{j\omega(L_1L_2 - M^2)} \\ \frac{-M}{j\omega(L_1L_2 - M^2)} & \frac{L_1}{j\omega(L_1L_2 - M^2)} \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j\omega(L_a + L_c) & j\omega L_c \\ j\omega L_c & j\omega(L_b + L_c) \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$L_a = L_1 - M$, $\quad L_b = L_2 - M$, $\quad L_c = M$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{j\omega L_A} + \frac{1}{j\omega L_C} & -\frac{1}{j\omega L_C} \\ -\frac{1}{j\omega L_C} & \frac{1}{j\omega L_B} + \frac{1}{j\omega L_C} \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$L_A = \frac{L_1L_2 - M^2}{L_2 - M}$, $\quad L_B = \frac{L_1L_2 - M^2}{L_1 - M}$

$L_C = \frac{L_1L_2 - M^2}{M}$

$\frac{V_2}{V_1} = \frac{N_2}{N_1} = n$, $\frac{I_2}{I_1} = \frac{N_1}{N_2} = \frac{1}{n}$, $\frac{V_2}{V_1} = -\frac{N_2}{N_1} = -n$, $\frac{I_2}{I_1} = -\frac{N_1}{N_2} = -\frac{1}{n}$

$Z_{in} = \frac{V_1}{I_1} = \frac{1}{n^2}\frac{V_2}{I_2}$ $\quad Z_{in} = \frac{Z_L}{n^2}$

$\frac{V_1}{V_2} = \frac{N_1 + N_2}{N_2} = \frac{I_2}{I_1}$

$\frac{V_1}{V_2} = \frac{N_1}{N_1 + N_2} = -\frac{I_2}{I_1}$

not offer isolation!

$S_1 = V_1 I_1^* = V_2 I_2^* = S_2$

## Frequency response

$H(\omega)$ = Voltage gain $= \frac{V_o(\omega)}{V_i(\omega)}$

$H(\omega)$ = Current gain $= \frac{I_o(\omega)}{I_i(\omega)}$

$H(\omega)$ = Transfer Impedance $= \frac{V_o(\omega)}{I_i(\omega)}$

$H(\omega)$ = Transfer Admittance $= \frac{I_o(\omega)}{V_i(\omega)}$

$H(\omega) = \frac{V_o}{V_i} = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1 + j\omega RC}$

$H = \frac{1}{\sqrt{1 + (\omega/\omega_0)^2}}$, $\quad \phi = -\tan^{-1}\frac{\omega}{\omega_0}$, $\quad \omega_0 = 1/RC$

| $\omega/\omega_0$ | $H$ | $\phi$ | $\omega/\omega_0$ | $H$ | $\phi$ |
|---|---|---|---|---|---|
| 0 | 1 | 0 | 10 | 0.1 | −84° |
| 1 | 0.71 | −45° | 20 | 0.05 | −87° |
| 2 | 0.45 | −63° | 100 | 0.01 | −89° |
| 3 | 0.32 | −72° | ∞ | 0 | −90° |

$G_{dB} = 10\log_{10}\frac{P_2}{P_1}$

$Z = H(\omega) = \frac{V_s}{I} = R + j\omega L + \frac{1}{j\omega C}$

$Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$

$\mathrm{Im}(Z) = \omega L - \frac{1}{\omega C} = 0$ 之谐振

$\omega_0 = \frac{1}{\sqrt{LC}}$ rad/s $\quad f_0 = \frac{1}{2\pi\sqrt{LC}}$ Hz

$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$ 本质相同

$|V_L| = \frac{V_m}{R}\omega_0 L = QV_m$

$|V_C| = \frac{V_m}{R}\frac{1}{\omega_0 C} = QV_m$

$Y = H(\omega) = \frac{I}{V} = \frac{1}{R} + j\omega C + \frac{1}{j\omega L}$

$Y = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$

| Characteristic | Series circuit | Parallel circuit |
|---|---|---|
| Resonant frequency, $\omega_0$ | $\frac{1}{\sqrt{LC}}$ | $\frac{1}{\sqrt{LC}}$ |
| Quality factor, Q | $\frac{\omega_0 L}{R}$ or $\frac{1}{\omega_0 RC}$ | $\frac{R}{\omega_0 L}$ or $\omega_0 RC$ |
| Bandwidth, B | $\frac{\omega_0}{Q}$ | $\frac{\omega_0}{Q}$ |
| Half-power frequencies, $\omega_1, \omega_2$ | $\omega_0\sqrt{1 + \left(\frac{1}{2Q}\right)^2} \mp \frac{\omega_0}{2Q}$ | $\omega_0\sqrt{1 + \left(\frac{1}{2Q}\right)^2} \mp \frac{\omega_0}{2Q}$ |
| For $Q \geq 10$, $\omega_1, \omega_2$ | $\omega_0 \mp \frac{B}{2}$ | $\omega_0 \mp \frac{B}{2}$ |

$H(\omega) = \frac{V_o}{V_i} = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1 + j\omega RC}$

$H(\omega_c) = \frac{1}{\sqrt{1 + \omega_c^2 R^2 C^2}} = \frac{1}{\sqrt{2}}$ $\quad \omega_c = \frac{1}{RC}$

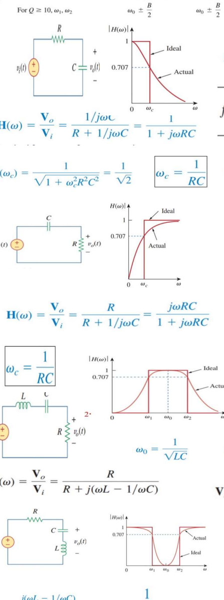
$H(\omega) = \frac{V_o}{V_i} = \frac{R}{R + 1/j\omega C} = \frac{j\omega RC}{1 + j\omega RC}$

$\omega_c = \frac{1}{RC}$

$H(\omega) = \frac{V_o}{V_i} = \frac{R}{R + j(\omega L - 1/\omega C)}$

$\omega_0 = \frac{1}{\sqrt{LC}}$

$H(\omega) = \frac{V_o}{V_i} = \frac{j(\omega L - 1/\omega C)}{R + j(\omega L - 1/\omega C)}$ $\quad \omega_0 = \frac{1}{\sqrt{LC}}$



## Fourier series

$\int_0^T \sin n\omega_0 t\cos m\omega_0 t\, dt = 0$

$\int_0^T \sin n\omega_0 t\sin m\omega_0 t\, dt = 0$, $(m \neq n)$

$\int_0^T \cos n\omega_0 t\cos m\omega_0 t\, dt = 0$, $(m \neq n)$

$\int_0^T \sin n\omega_0 t\, dt = 0$

$\int_0^T \cos n\omega_0 t\, dt = 0$

$\int_0^T \sin^2 n\omega_0 t\, dt = \frac{T}{2}$

$\int_0^T \cos^2 n\omega_0 t\, dt = \frac{T}{2}$

$f(t) = a_0 + \sum_{n=1}^{\infty} A_n\cos(n\omega_0 t + \phi_n)$

$A_n = \sqrt{a_n^2 + b_n^2}$, $\quad \phi_n = -\tan^{-1}\frac{b_n}{a_n}$

$f(t) = \frac{1}{2} + \frac{2}{\pi}\sum_{k=1}^{\infty}\frac{1}{n}\sin n\pi t$, $\quad n = 2k - 1$

$f(t) = \frac{4A}{\pi}\sum_{n=1}^{\infty}\frac{1}{2n - 1}\sin(2n - 1)\omega_0 t$

$f(t) = \frac{A\tau}{T} + \frac{2A}{T}\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi\tau}{T}\cos n\omega_0 t$

$f(t) = \frac{A}{2} - \frac{A}{\pi}\sum_{n=1}^{\infty}\frac{\sin n\omega_0 t}{n}$

4. Triangular wave

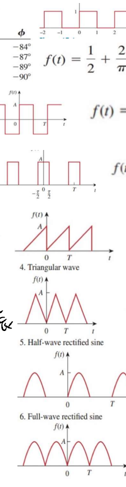
$f(t) = \frac{A}{2} - \frac{4A}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{(2n - 1)^2}\cos(2n - 1)\omega_0 t$

5. Half-wave rectified sine

$f(t) = \frac{A}{\pi} + \frac{A}{2}\sin\omega_0 t - \frac{2A}{\pi}\sum_{n=1}^{\infty}\frac{1}{4n^2 - 1}\cos 2n\omega_0 t$

6. Full-wave rectified sine

$f(t) = \frac{2A}{\pi} - \frac{4A}{\pi}\sum_{n=1}^{\infty}\frac{1}{4n^2 - 1}\cos n\omega_0 t$



$v(t) = V_{dc} + \sum_{n=1}^{\infty} V_n\cos(n\omega_0 t - \theta_n)$

$i(t) = I_{dc} + \sum_{m=1}^{\infty} I_m\cos(m\omega_0 t - \phi_m)$

$P = \frac{1}{T}\int_0^T vi\, dt$

$P = V_{dc}I_{dc} + \frac{1}{2}\sum_{n=1}^{\infty} V_n I_n\cos(\theta_n - \phi_n)$

$P = V_{rms}I_{rms}\cos(\theta_v - \theta_i) = S\cos(\theta_v - \theta_i)$ $\quad S = V_{rms}I_{rms}$

$F_{rms} = \sqrt{a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty} A_n^2}$ $\quad P = \frac{F_{rms}^2}{R}$

$F_{rms} = \sqrt{a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2)}$ $\quad P = RF_{rms}^2$

$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$ $\quad c_n = \frac{1}{T}\int_0^T f(t)e^{-jn\omega_0 t}dt$

$c_0 = a_0$, $\quad c_n = \frac{(a_n - jb_n)}{2}$, $\quad c_{-n} = c_n^* = \frac{(a_n + jb_n)}{2}$

Figure 17.20 For Example 17.6.

magnetically coupled: $M_1 = M_{21} = M$ reverse

Figure 17.1 For Example 17.1; a square wave.

上一页表

$v_s(t) = \frac{1}{2} + \frac{2}{\pi}\sum_{k=1}^{\infty}\frac{1}{n}\sin n\pi t$, $\quad n = 2k - 1$

$V_o = \frac{j\omega_n L}{R + j\omega_n L}V_s = \frac{j2n\pi}{5 + j2n\pi}V_s$

$V_s = \frac{1}{2}$ $\Rightarrow$ $V_o = 0$

$V_s = \frac{2}{n\pi}\angle -90°$ $\quad V_o = \frac{2n\pi\angle 90°}{\sqrt{25 + 4n^2\pi^2}\angle\tan^{-1}2n\pi/5}\left(\frac{2}{n\pi}\angle -90°\right)$

$= \frac{4\angle -\tan^{-1}2n\pi/5}{\sqrt{25 + 4n^2\pi^2}}$

$v_o(t) = \sum_{k=1}^{\infty}\frac{4}{\sqrt{25 + 4n^2\pi^2}}\cos\left(n\pi t - \tan^{-1}\frac{2n\pi}{5}\right)$, $\quad n = 2k - 1$

$f(t) = a_0 + \sum_{n=1}^{\infty}(a_n\cos n\omega_0 t + b_n\sin n\omega_0 t)$ (dc, ac)

$a_0 = \frac{1}{T}\int_0^T f(t)\, dt$ $\quad a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega_0 t\, dt$ $\quad b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega_0 t\, dt$

## Fourier transform

$F(\omega) = \mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$

$f(t) = \mathcal{F}^{-1}[F(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}d\omega$

$\mathcal{F}[f(at)] = \frac{1}{|a|}F\left(\frac{\omega}{a}\right)$ $\quad \mathcal{F}[f(t - t_0)] = e^{-j\omega t_0}F(\omega)$

$\mathcal{F}[f(t)e^{j\omega_0 t}] = F(\omega - \omega_0)$

$\mathcal{F}[f'(t)] = j\omega F(\omega)$

$\mathcal{F}[f^{(n)}(t)] = (j\omega)^n F(\omega)$

$\mathcal{F}\left[\int_{-\infty}^{t} f(t)\, dt\right] = \frac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega)$

$\mathcal{F}[f(-t)] = F(-\omega) = F^*(\omega)$

$\mathcal{F}[f(t)] = F(\omega)$ $\Rightarrow$ $\mathcal{F}[F(t)] = 2\pi f(-\omega)$

$f(t) = e^{-|t|}$, $\quad F(\omega) = \frac{2}{\omega^2 + 1}$

$F(t) = 2/(t^2 + 1)$ $\quad 2\pi f(\omega) = 2\pi e^{-|\omega|}$

| $f(t)$ | $F(\omega)$ |
|---|---|
| $\delta(t)$ | $1$ |
| $1$ | $2\pi\delta(\omega)$ |
| $u(t)$ | $\pi\delta(\omega) + \frac{1}{j\omega}$ |
| $u(t + \tau) - u(t - \tau)$ | $2\frac{\sin\omega\tau}{\omega}$ |
| $\|t\|$ | $\frac{-2}{\omega^2}$ |
| $\mathrm{sgn}(t)$ | $\frac{2}{j\omega}$ |
| $e^{-at}u(t)$ | $\frac{1}{a + j\omega}$ |
| $e^{at}u(-t)$ | $\frac{1}{a - j\omega}$ |
| $t^n e^{-at}u(t)$ | $\frac{n!}{(a + j\omega)^{n+1}}$ |
| $e^{-a\|t\|}$ | $\frac{2a}{a^2 + \omega^2}$ |
| $e^{j\omega_0 t}$ | $2\pi\delta(\omega - \omega_0)$ |
| $\sin\omega_0 t$ | $j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$ |
| $\cos\omega_0 t$ | $\pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$ |
| $e^{-at}\sin\omega_0 t u(t)$ | $\frac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$ |
| $e^{-at}\cos\omega_0 t u(t)$ | $\frac{a + j\omega}{(a + j\omega)^2 + \omega_0^2}$ |

The instantaneous powers in the three phases:

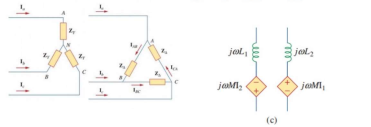
$p = p_a + p_b + p_c = v_{AN}i_a + v_{BN}i_b + v_{CN}i_c = 3V_pI_p\cos\theta = \sqrt{3}V_LI_L\cos$

The average powers per phase: $P_p = V_pI_p\cos\theta$

The reactive powers per phase: $Q_p = V_pI_p\sin\theta$

The apparent powers per phase: $S_p = V_pI_p$

The complex powers per phase: $S_p = P_p + jQ_p = V_pI_p^*$



$L = L_1 + L_2 + 2M$

$L = L_1 + L_2 - 2M$

For coil 1, KVL gives,

$V = (Z_1 + j\omega L_1)I_1 - j\omega MI_2$

For coil 2, KVL gives,

$0 = -j\omega MI_1 + (Z_L + j\omega L_2)I_2$